Network Management, Monitoring and Automation

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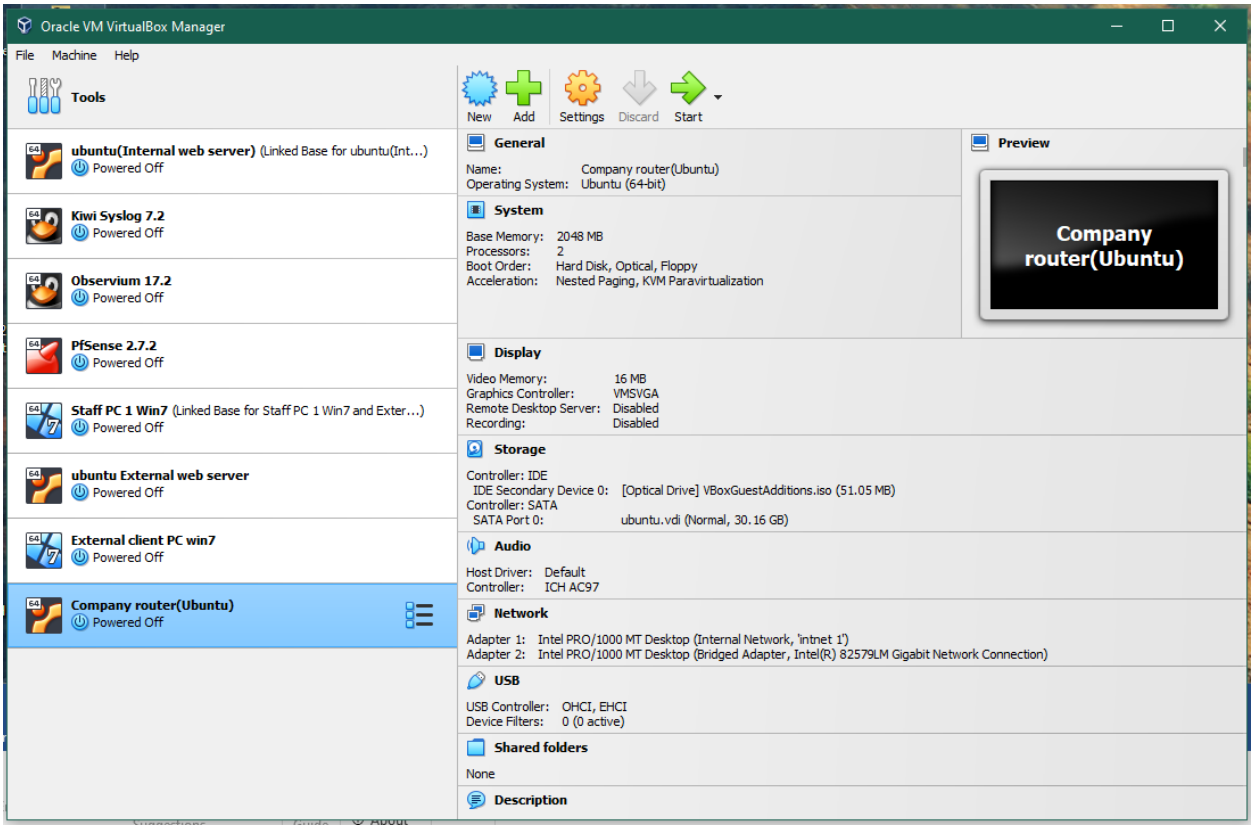
a working simulation of the following practical.

The first step taken was to install all the required operating systems in virtualbox as shown in the screenshot below.

For the internal web server, apache in Ubuntu was utilized and for the staff pc1, windows 7 was used. Observium and Kiwi syslog was installed in the virtualbox.



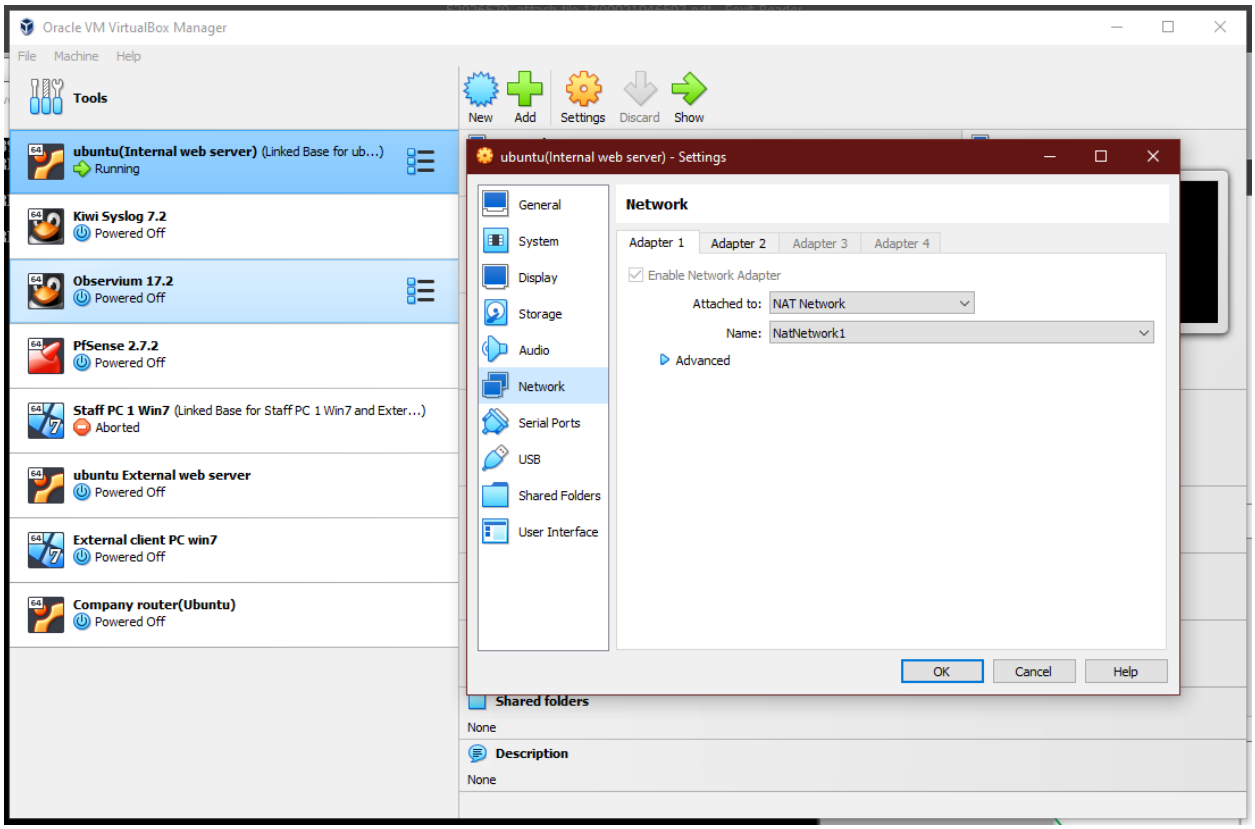
Having installed all the required machines, the network configurations were now set where the devices in the internal network were added to the Nat network1 network that included the internal web server, staff PC 1, Observium, Kiwi Syslog, and PfSense.

Specific configurations.

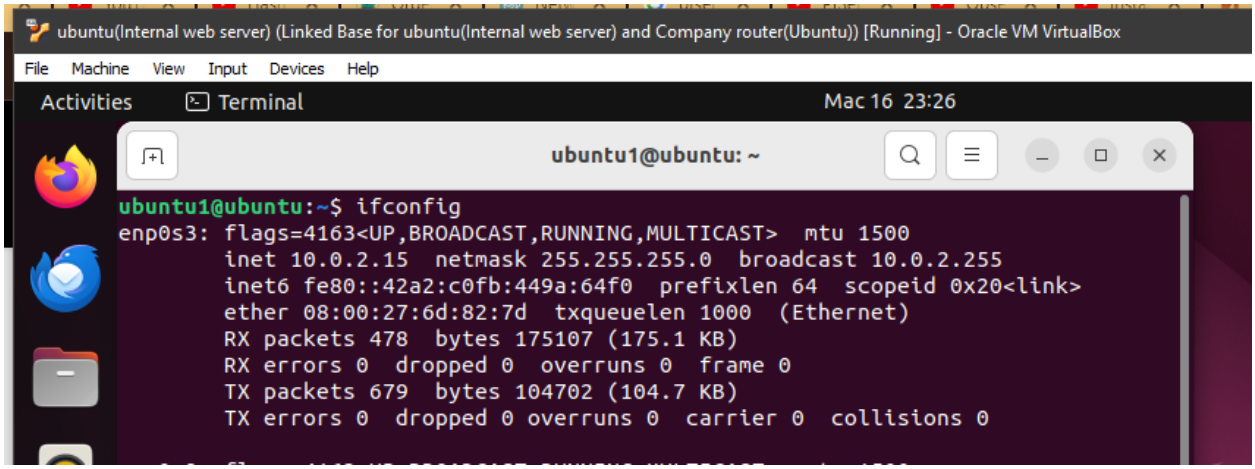
Internal network.

In the first configurations, we had to ensure that all the devices in the internal network and the external client pc and the external Web server were communicating. To achieve this, the networks were configured in virtual box as shown in the following section below.

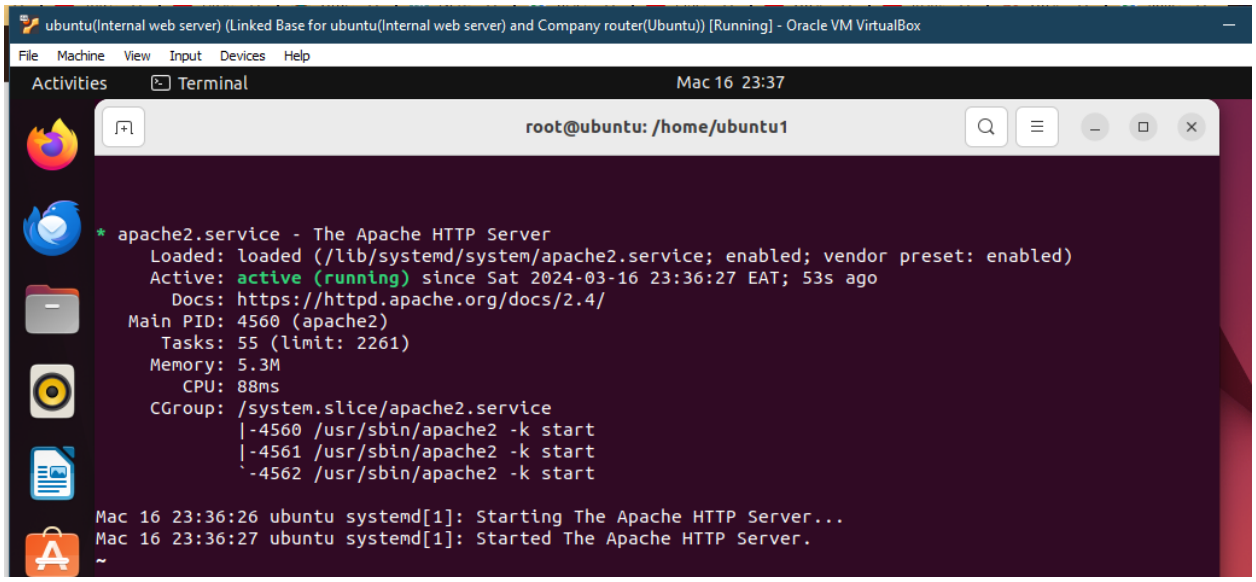
1. Virtualbox networks
    1. Internal Web Server



After starting, an automatic DHCP IP address was assigned which is 10.0.2.15 as shown below.

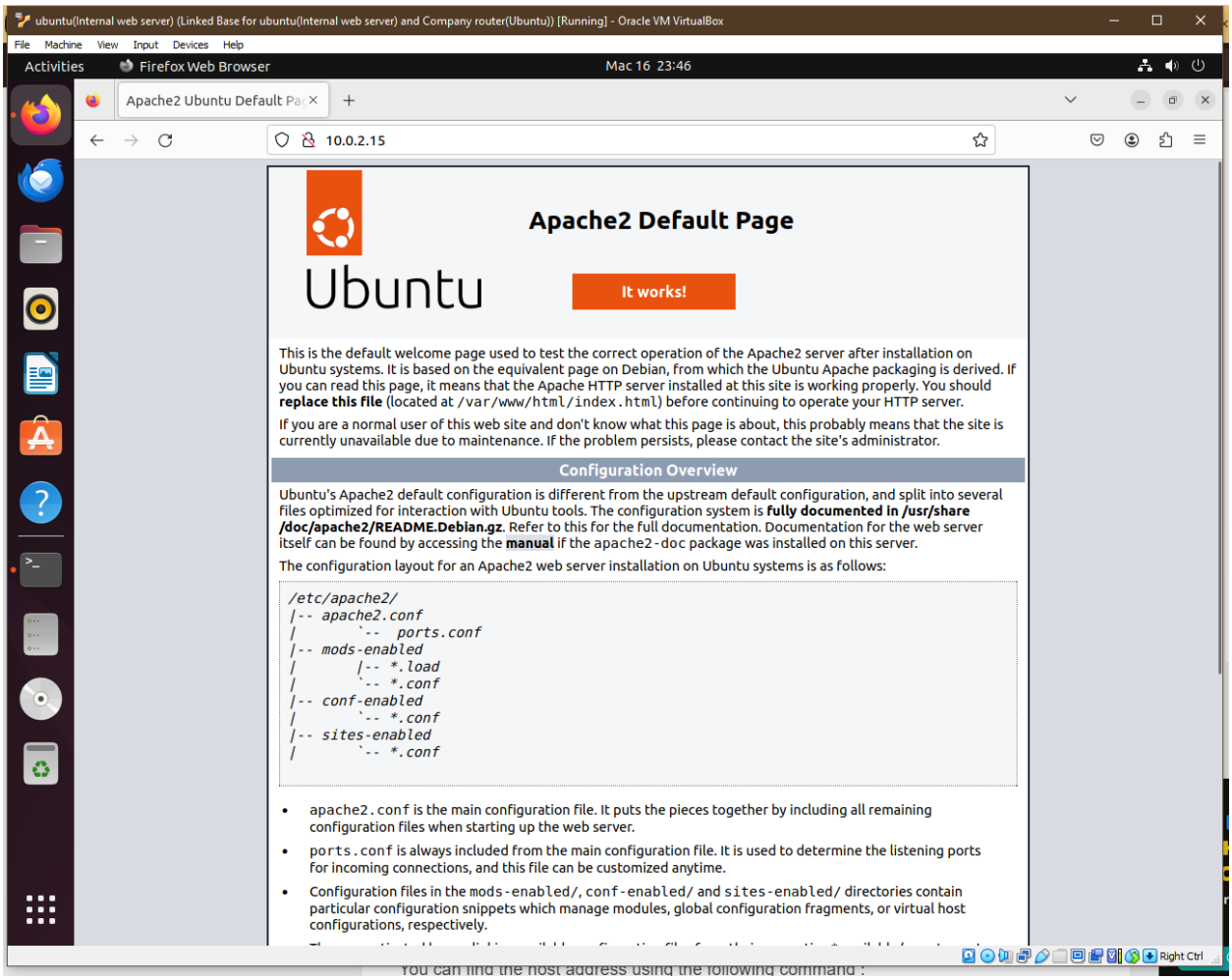


Being the internal server, an Apache server was downloaded and installed to run on the internal Ubuntu server.
Apache server was then started as shown in the following screenshot.



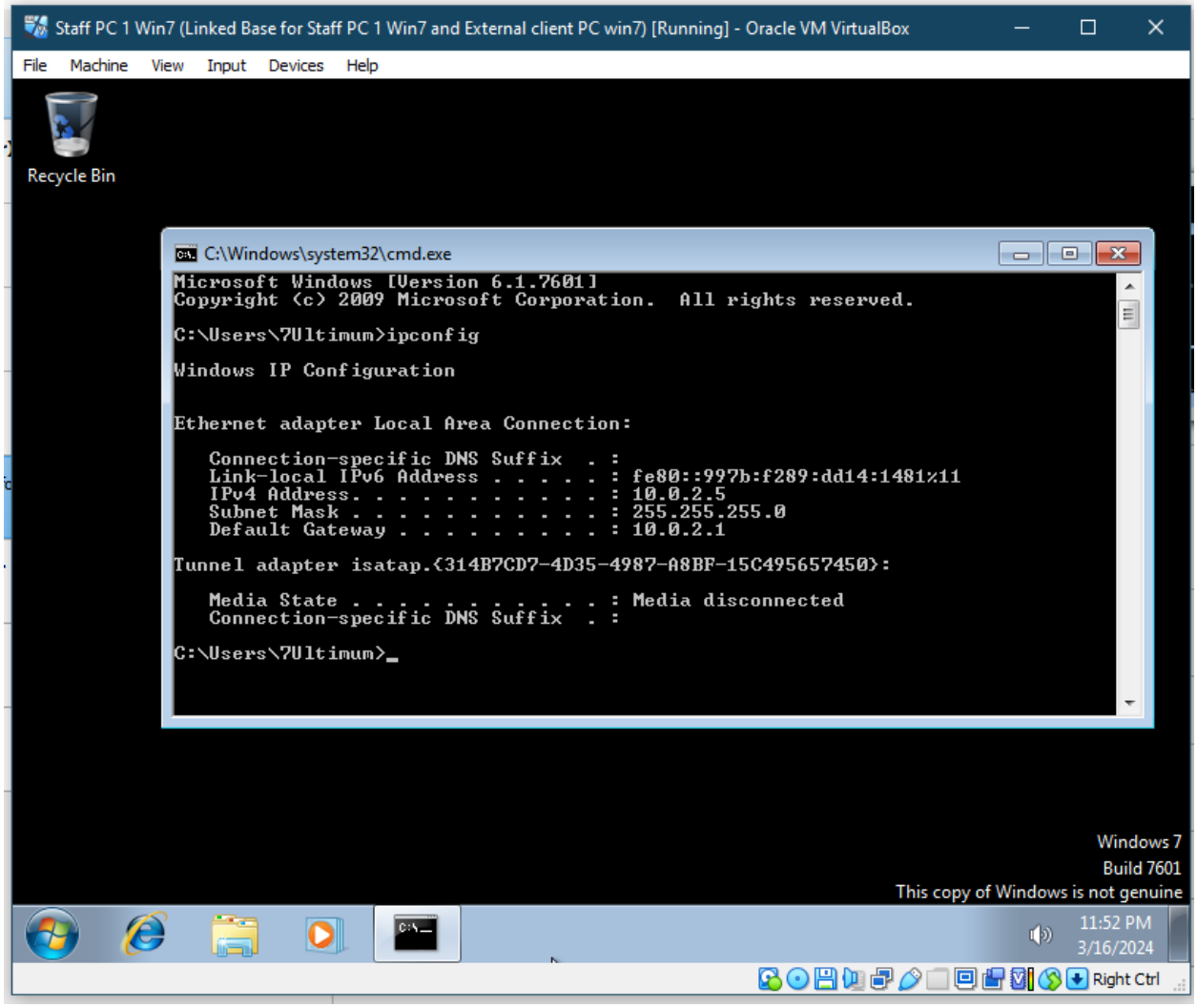
The next step involved configuring the firewall settings for Apache Server

The firewall was set to allow all communications and tested on the web servers webpage as shown here

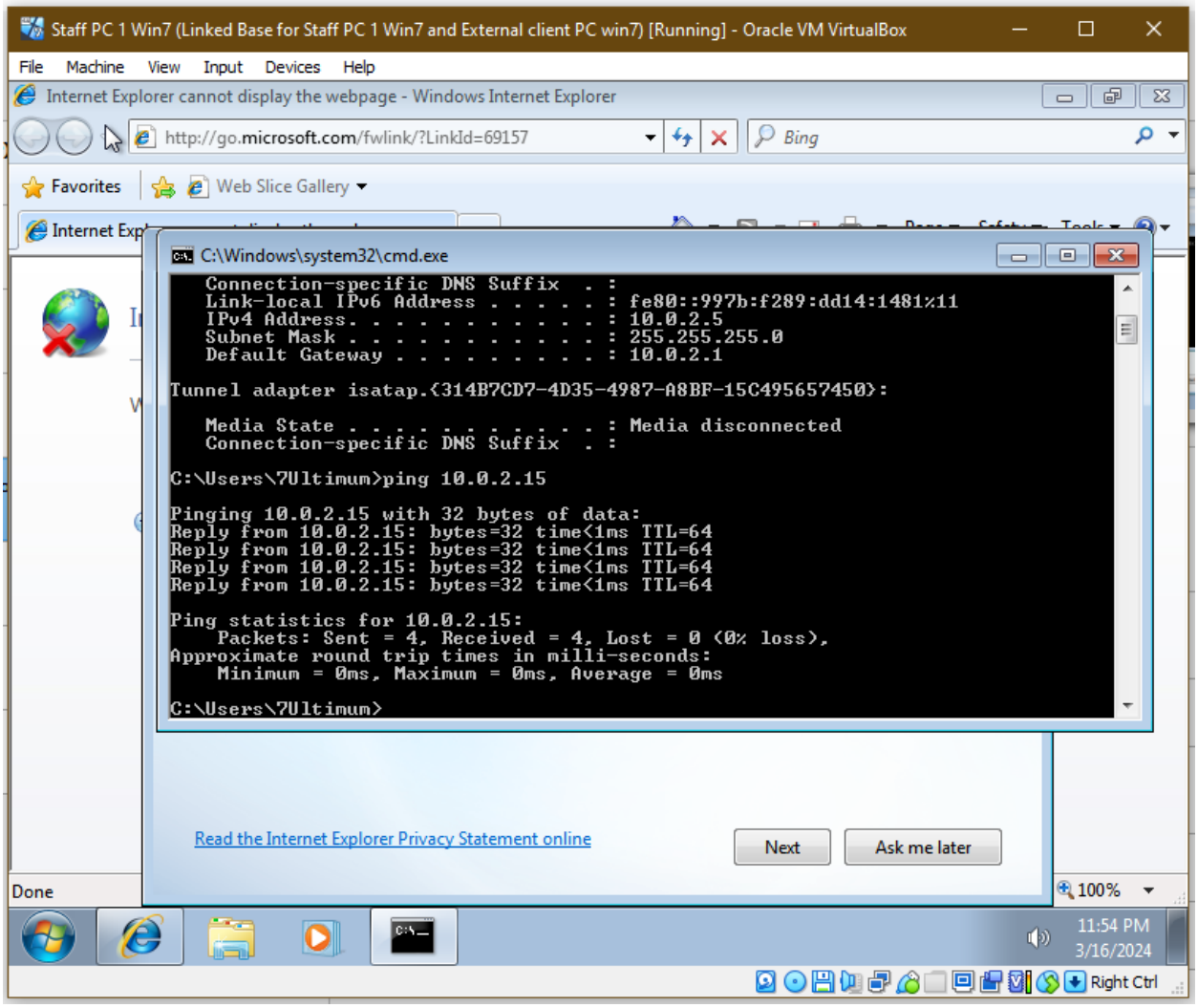


2. Staff PC 1

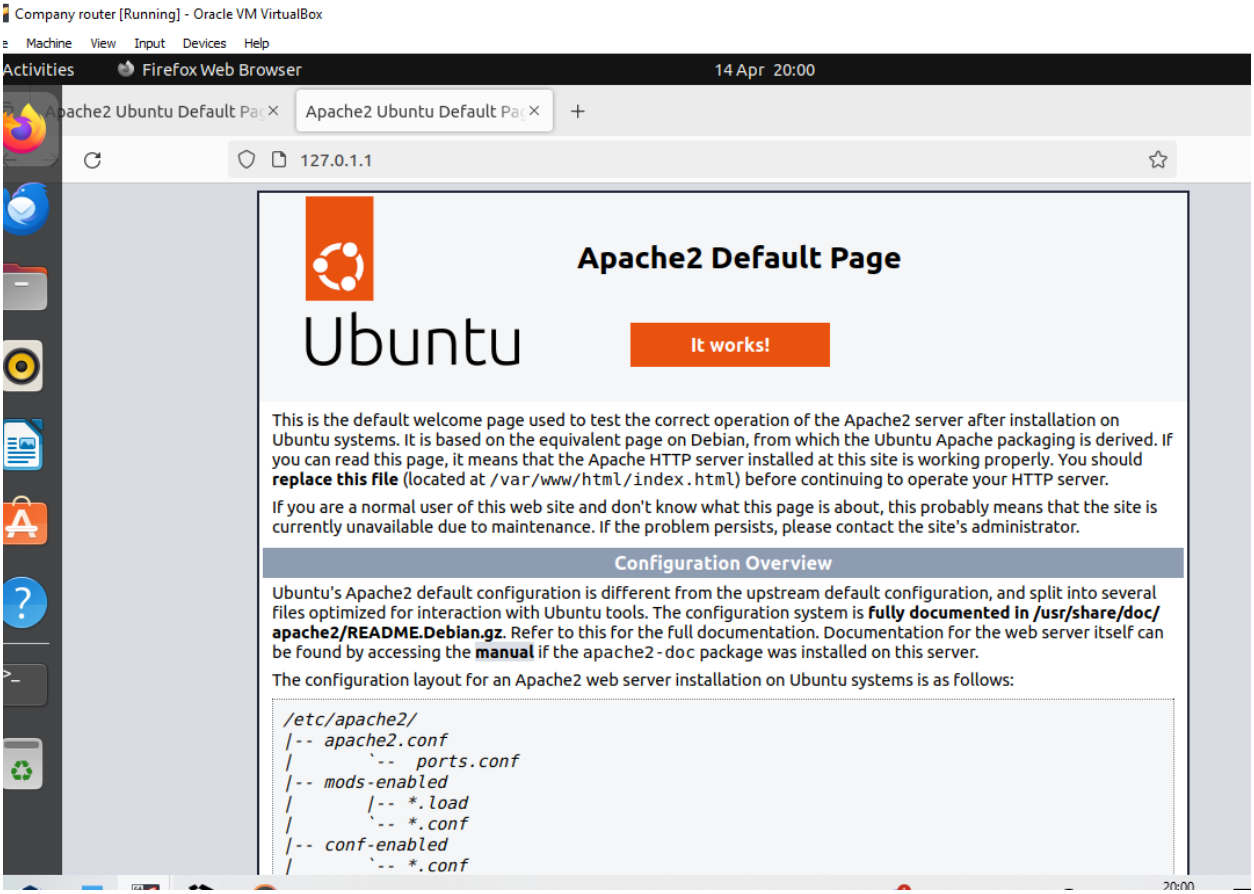
The staff PC1 was utilizing a windows 7 operating system. The windows was also configured in the internal network hence connected to the Natnetwork 1 interface.



Now PC1 was tested whether it can communicate with the Internal web server by pinging its address and the result indicated that successful communication was established.

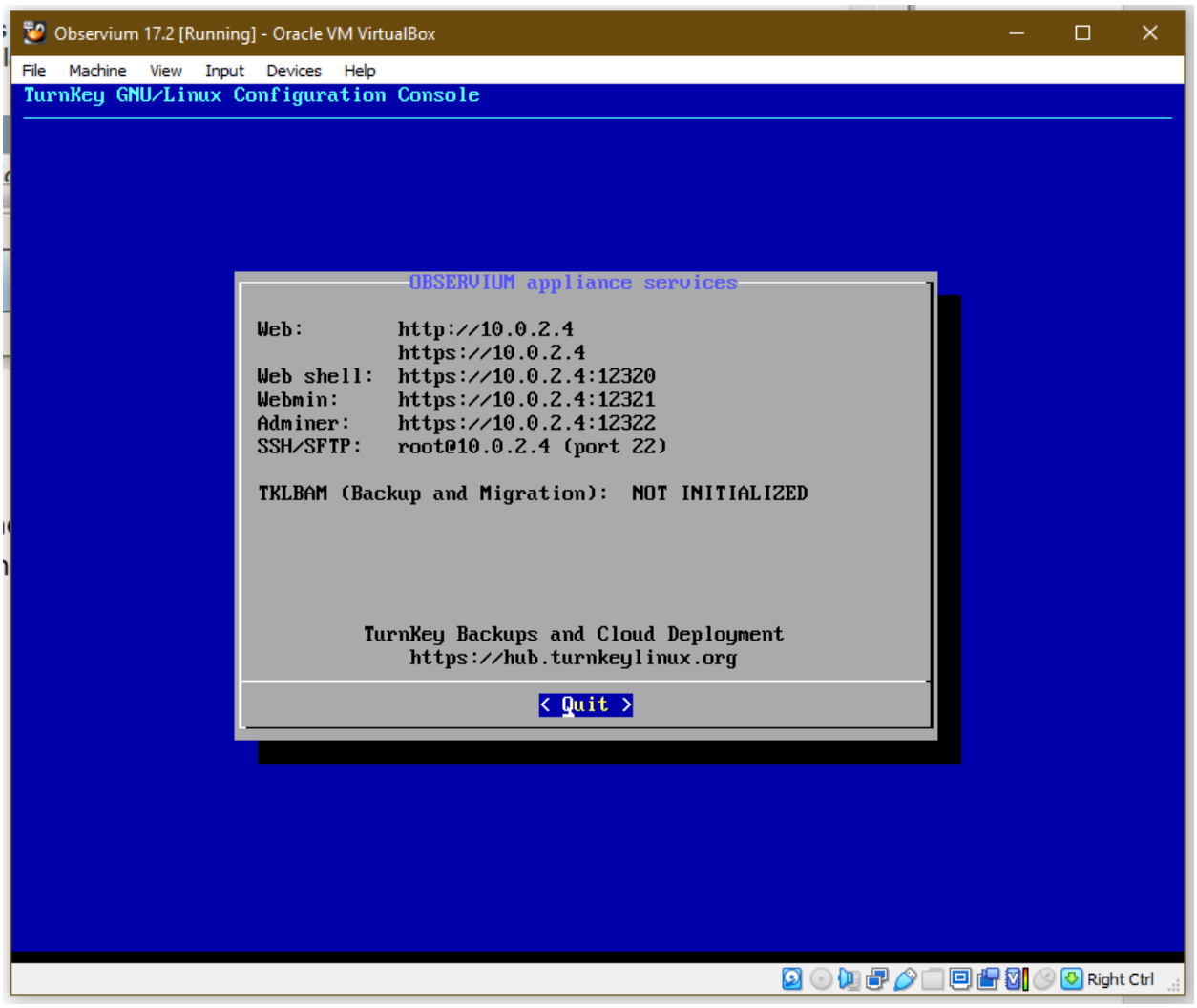


The pc1 client was also able to reach the internal apache web server as shown below.



3. Observium

The very first step was to install the observium and start it in virtualbox. Upon starting, the connection links were provided as shown in the following screenshot.



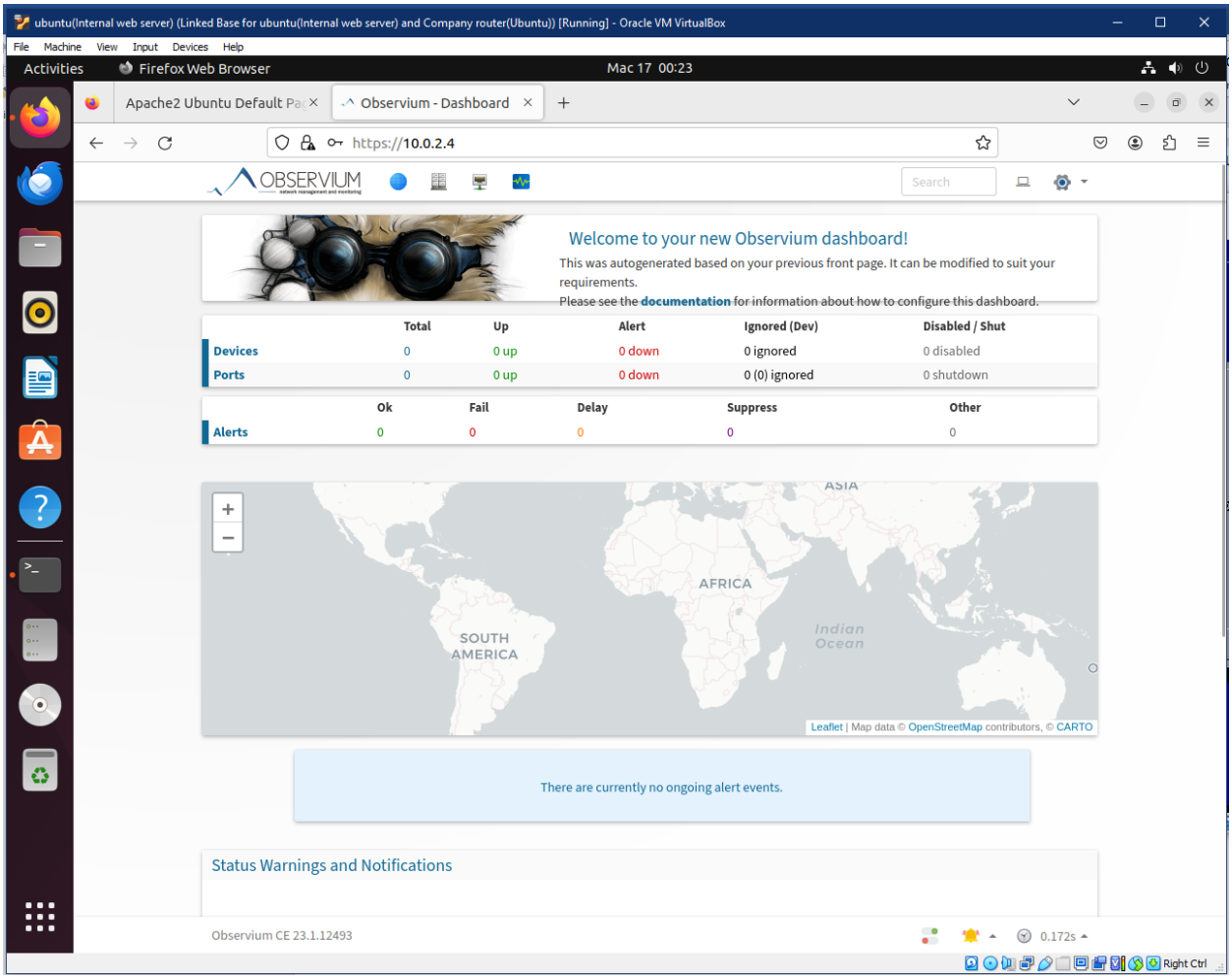
After accessing it via the web browser in the internal web server, one could now login using the credentials set up during installation. This successfully loged in as shown here

Observium 17.2 [Running] - Oracle VM VirtualBox

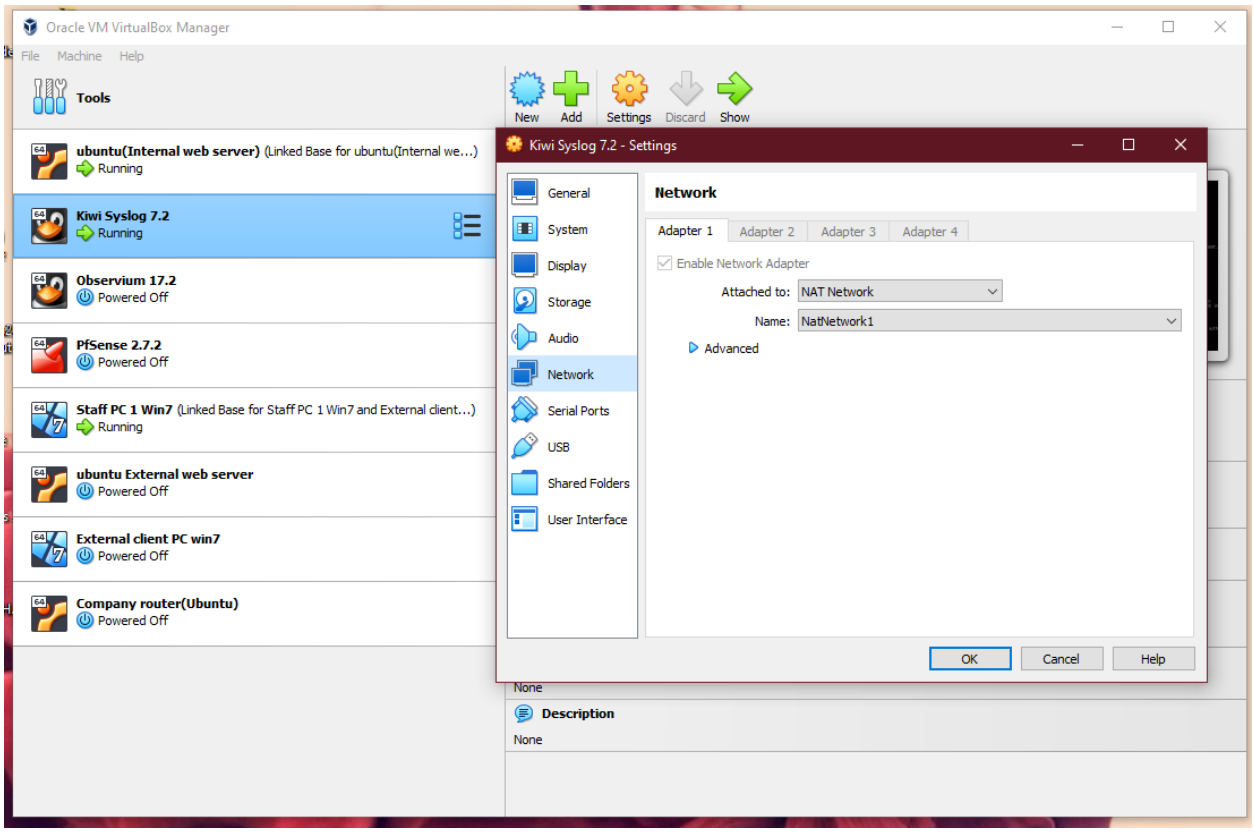
File Machine View Input Devices Help

```
Debian GNU/Linux 11 observium tty1

observium login: admin
Password:

Login incorrect
observium login: root
Password:
Welcome to Observium, TurnKey GNU/Linux 17.2 (Debian 11/Bullseye)

  System information for Sat Mar 16 21:09:31 2024 (UTC+0000)

    System load:  0.08              Memory usage:  19.5%
    Processes:    97                Swap usage:    0.0%
    Usage of /:   43.3% of 5.34GB   IP address for eth0: 10.0.2.4

  TKLBAM (Backup and Migration):  NOT INITIALIZED

    To initialize TKLBAM, run the "tklbam-init" command to link this
    system to your TurnKey Hub account. For details see the man page or
    go to:

        https://www.turnkeylinux.org/tklbam


    For Advanced commandline config run:    confconsole

  For more info see: https://www.turnkeylinux.org/docs/confconsole

Linux observium 5.10.0-21-amd64 #1 SMP Debian 5.10.162-1 (2023-01-21) x86_64
Last login: Sat Mar 16 17:21:47 UTC 2024 on tty1
root@observium ~#
```

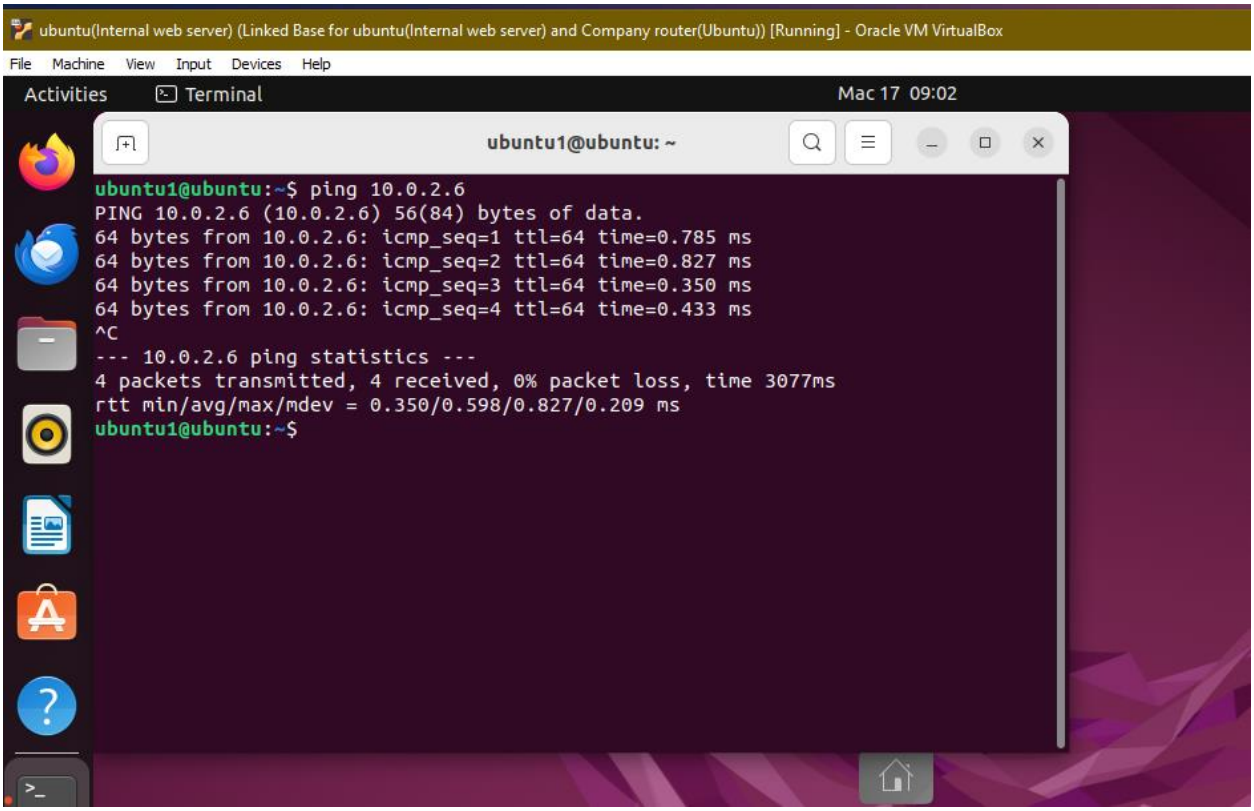
Right Ctrl

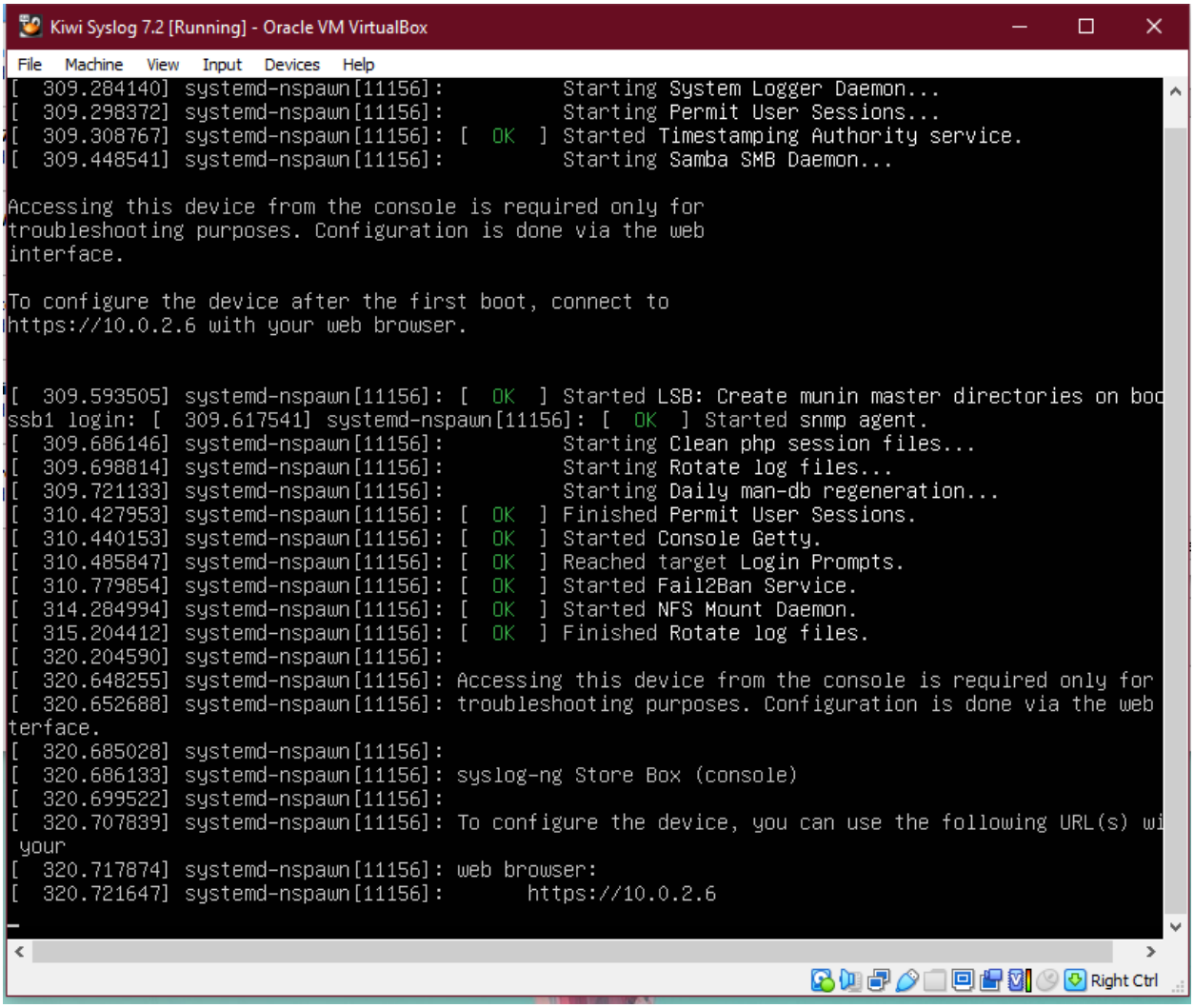


4. Kiwi Syslog
   Being in the internal network, Kiwi Syslog was configure by connecting its Ethernet to the Natnetwork1 for internal network.



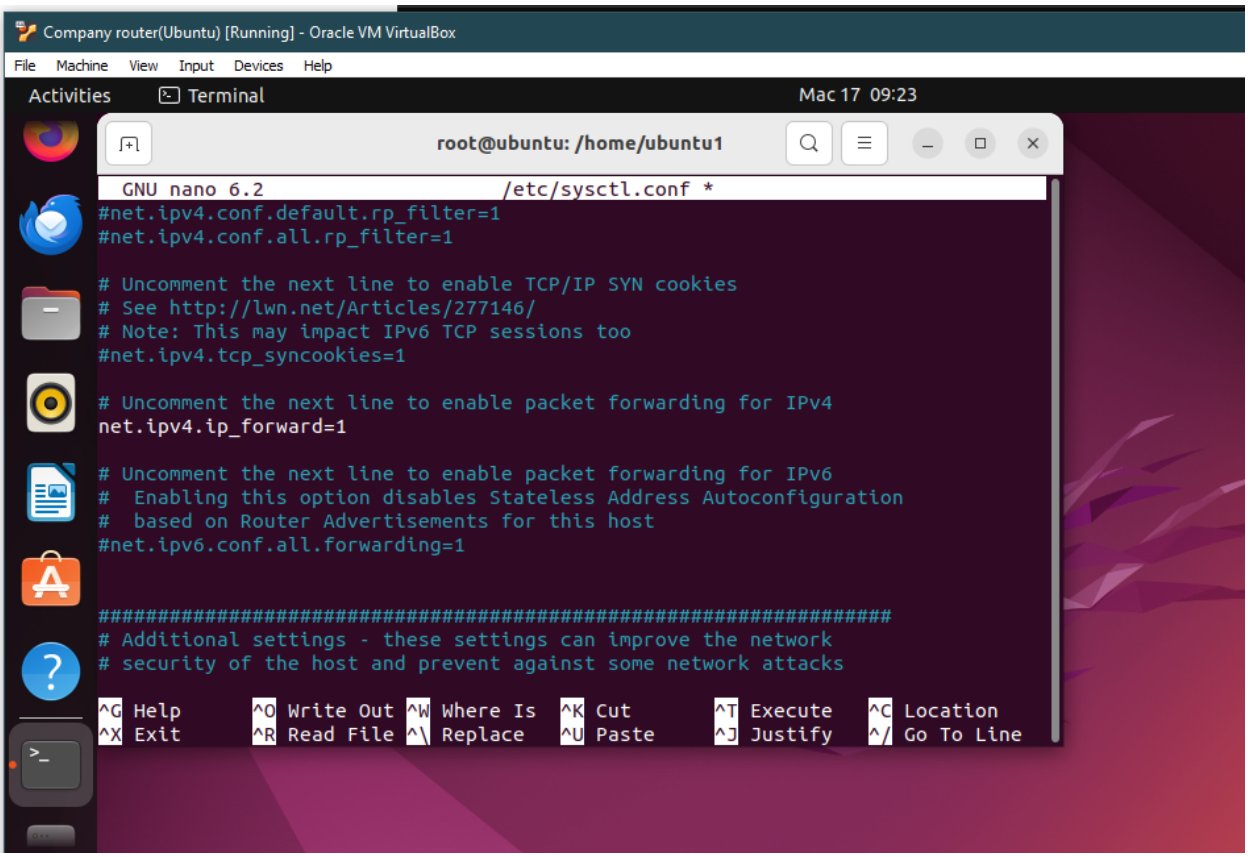
After running it, the machine started and was now accessible as we tested that the internal server was able to communicate with it as shown here,


ubuntu(Internal web server) (Linked Base for ubuntu(Internal web server) and Company router(Ubuntu)) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities
Terminal
Mac 17 09:02
ubuntu1@ubuntu: ~
ubuntu1@ubuntu:~$ ping 10.0.2.6
PING 10.0.2.6 (10.0.2.6) 56(84) bytes of data.
64 bytes from 10.0.2.6: icmp_seq=1 ttl=64 time=0.785 ms
64 bytes from 10.0.2.6: icmp_seq=2 ttl=64 time=0.827 ms
64 bytes from 10.0.2.6: icmp_seq=3 ttl=64 time=0.350 ms
64 bytes from 10.0.2.6: icmp_seq=4 ttl=64 time=0.433 ms
^C
--- 10.0.2.6 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3077ms
rtt min/avg/max/mdev = 0.350/0.598/0.827/0.209 ms
ubuntu1@ubuntu:~$



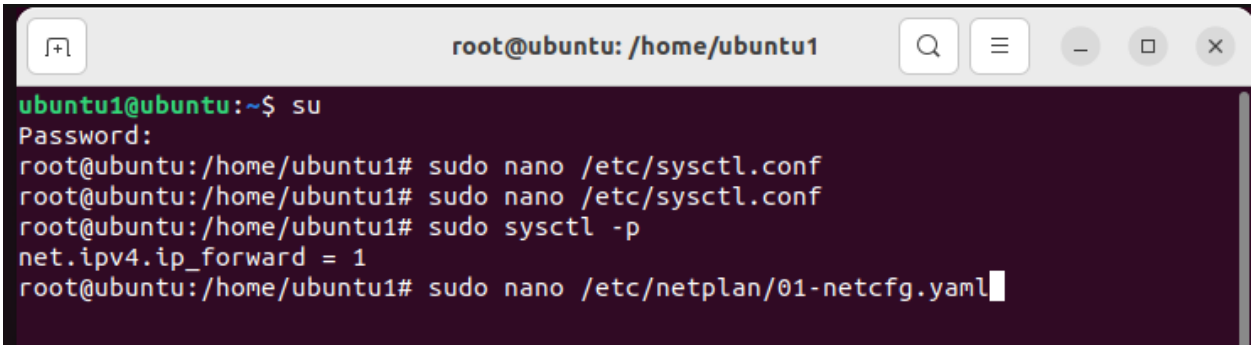
5. Company router

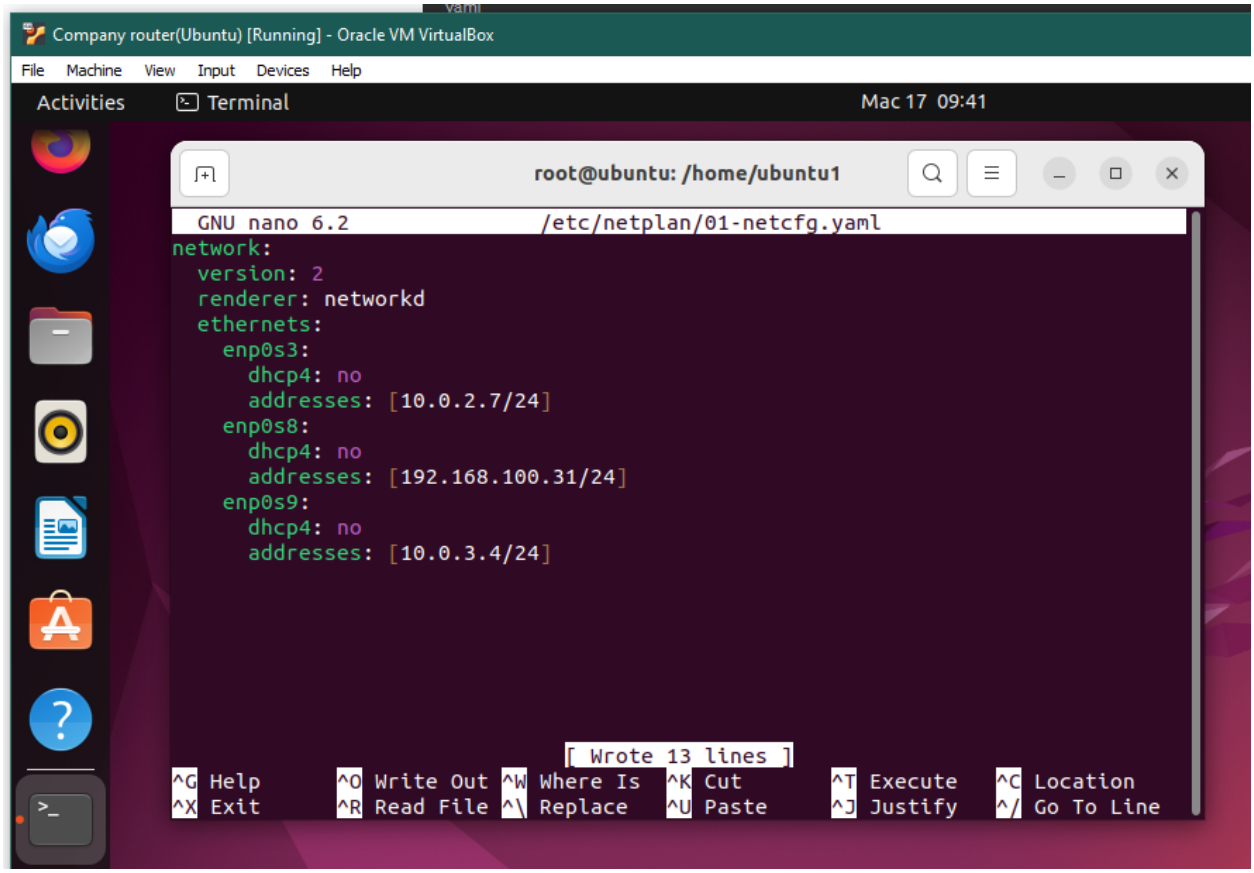
   For the company router, an instance of Ubuntu was used to bridge between the external devices network and the internal network devices as shown. The router was configured as shown in the images below. In this case the interfaces for both Natnetwork1 and Natnetwork2 were used.
   After starting the router had to be configure by enabling IP forwarding as shown here



This was done using the command **sudo nano /etc/sysctl.conf.** The line above was uncommented.

Then we open the 01-netcfg.yaml file and add configurations for each interface as shown here.





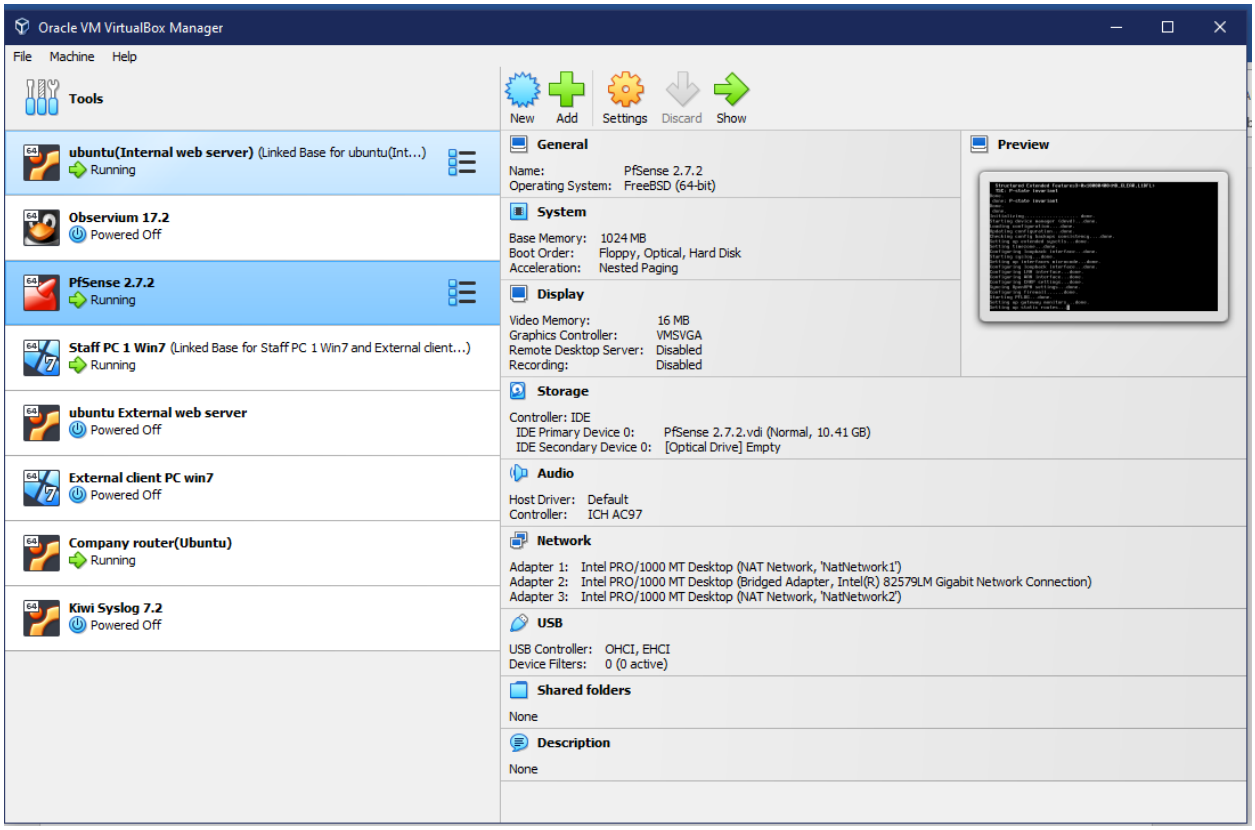
The configurations are then applied.

```
root@ubuntu:/home/ubuntu1# sudo nano /etc/netplan/01-netcfg.yaml
root@ubuntu:/home/ubuntu1# sudo netplan apply
root@ubuntu:/home/ubuntu1#
```

Now configure the IP tables to enable NAT forwarding by adding the routes.

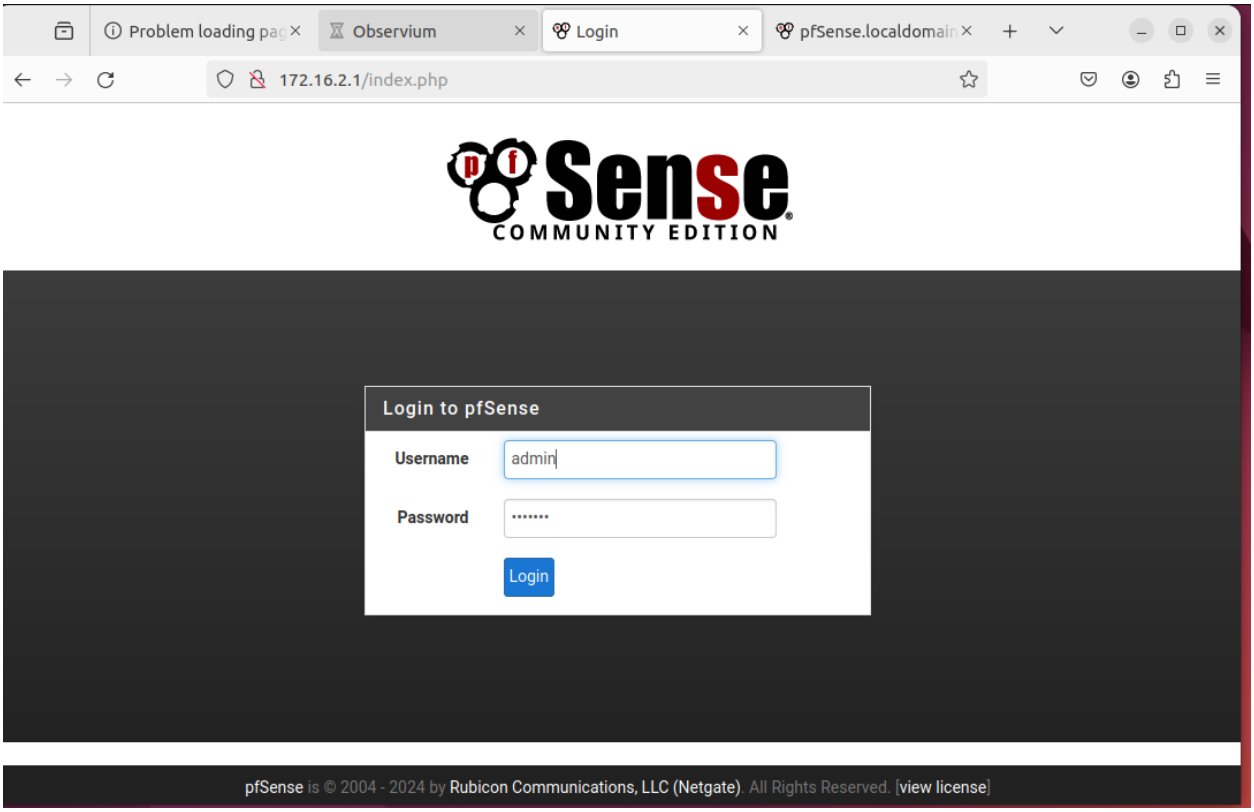
```
root@ubuntu:/home/ubuntu1# sudo iptables -t nat -A POSTROUTING -o enp0s8 -j MASQ
UERADE
root@ubuntu:/home/ubuntu1# sudo ip route add 10.0.2.0/24 via 10.0.2.7
root@ubuntu:/home/ubuntu1# sudo ip route add 10.0.3.0/24 via 10.0.3.4
root@ubuntu:/home/ubuntu1# sudo nano /etc/sysctl.conf
root@ubuntu:/home/ubuntu1# sudo systemctl restart networking
Failed to restart networking.service: Unit networking.service not found.
root@ubuntu:/home/ubuntu1# sudo systemctl restart systemd-networkd
root@ubuntu:/home/ubuntu1#
```

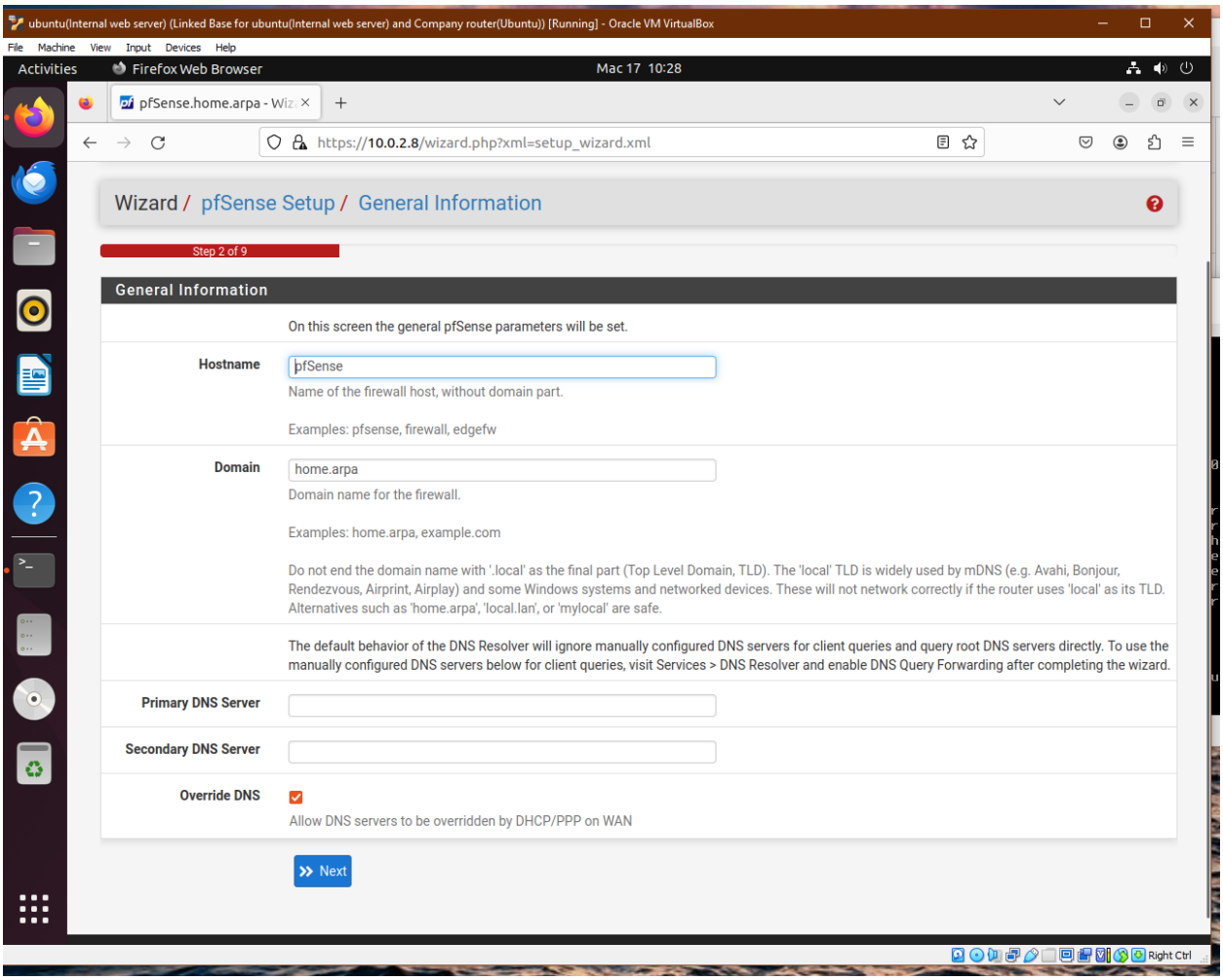
6. PfSense
   During installation, the pfsense network was given access to both networks as shown in the following image.



After successfully starting, the required configurations are made as shown below.

The next step is accessing the web interface at the provided link for additional configurations.


Problem loading pag
Observium
Login
pfSense.localdomain
172.16.2.1/index.php
pfSense
COMMUNITY EDITION
Login to pfSense
Username
admin
Password
Login
pfSense is © 2004 - 2024 by Rubicon Communications, LLC (Netgate). All Rights Reserved. [view license]


ubuntu(Internal web server) (Linked Base for ubuntu(Internal web server) and Company router(Ubuntu)) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities
Firefox Web Browser
Mac 17 10:28
pfSense.home.arpa - Wiz
https://10.0.2.8/wizard.php?xml=setup_wizard.xml
Wizard / pfSense Setup / General Information
Step 2 of 9
General Information
On this screen the general pfSense parameters will be set.
Hostname
pfSense
Name of the firewall host, without domain part.
Examples: pfsense, firewall, edgefw
Domain
home.arpa
Domain name for the firewall.
Examples: home.arpa, example.com
Do not end the domain name with '.local' as the final part (Top Level Domain, TLD). The 'local' TLD is widely used by mDNS (e.g. Avahi, Bonjour, Rendezvous, Airprint, Airplay) and some Windows systems and networked devices. These will not network correctly if the router uses 'local' as its TLD. Alternatives such as 'home.arpa', 'local.lan', or 'mylocal' are safe.
The default behavior of the DNS Resolver will ignore manually configured DNS servers for client queries and query root DNS servers directly. To use the manually configured DNS servers below for client queries, visit Services > DNS Resolver and enable DNS Query Forwarding after completing the wizard.
Primary DNS Server
Secondary DNS Server
Override DNS
Allow DNS servers to be overridden by DHCP/PPP on WAN
Next
Right Ctrl

ubuntu(Internal web server) (Linked Base for ubuntu(Internal web server) and Company router(Ubuntu)) [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Activities Firefox Web Browser Mac 17 10:30

pfSense.home.arpa - Wiz

https://10.0.2.8/wizard.php?xml=setup_wizard.xml

PPTP Password	
Show PPTP password	☐ Reveal password characters
PPTP Local IP Address	
pptplocalsubnet	32
PPTP Remote IP Address	
PPTP Dial on demand	☐ Enable Dial-On-Demand mode This option causes the interface to operate in dial-on-demand mode, allowing a virtual full time connection. The interface is configured, but the actual connection of the link is delayed until qualifying outgoing traffic is detected.
PPTP Idle timeout	If no qualifying outgoing packets are transmitted for the specified number of seconds, the connection is brought down. An idle timeout of zero disables this feature.

RFC1918 Networks

Block RFC1918 Private Networks	☑ Block private networks from entering via WAN When set, this option blocks traffic from IP addresses that are reserved for private networks as per RFC 1918 (10/8, 172.16/12, 192.168/16) as well as loopback addresses (127/8). This option should generally be left turned on, unless the WAN network lies in such a private address space, too.

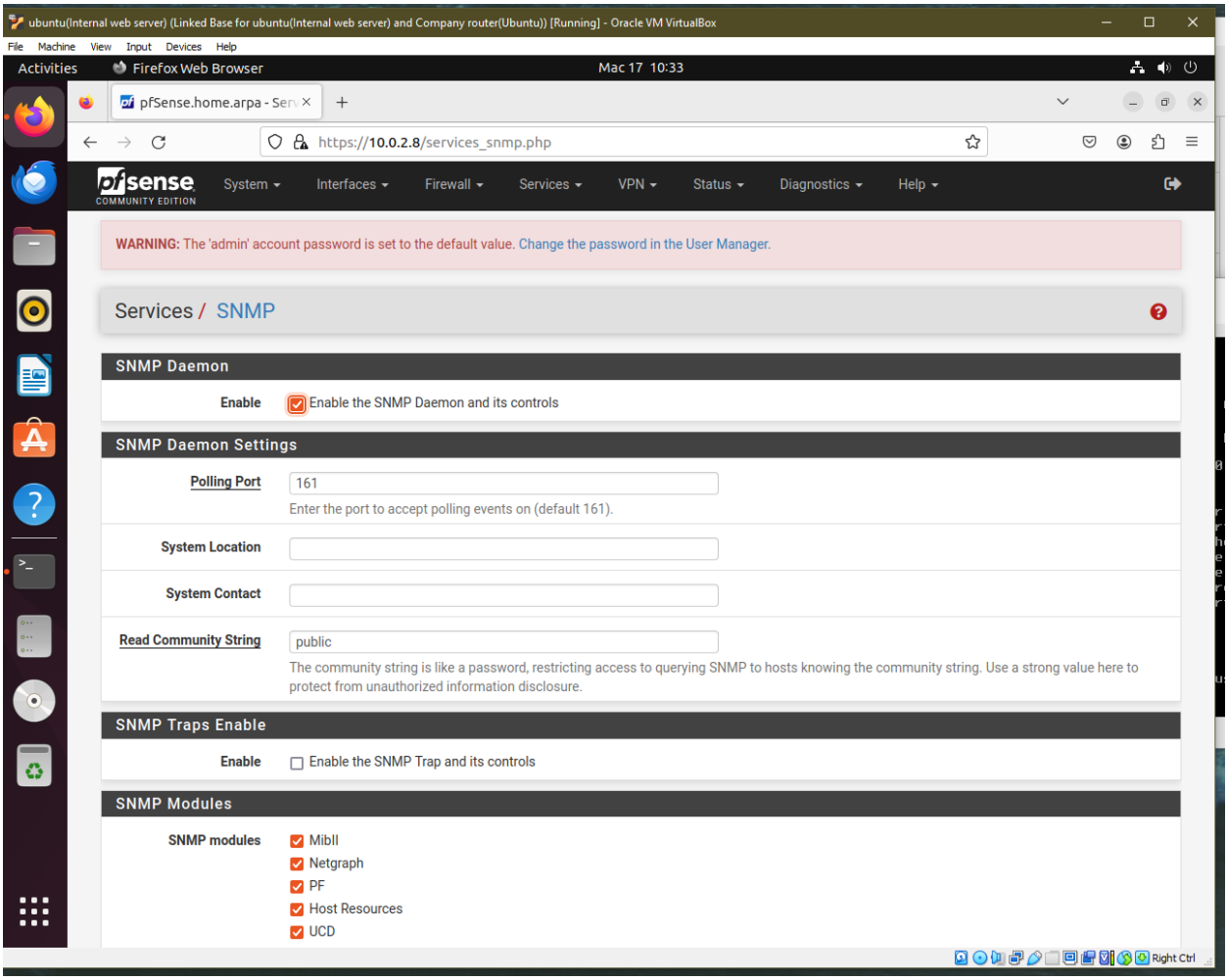
Block bogon networks

Block bogon networks	☑ Block non-Internet routed networks from entering via WAN When set, this option blocks traffic from IP addresses that are reserved (but not RFC 1918) or not yet assigned by IANA. Bogons are prefixes that should never appear in the Internet routing table, and obviously should not appear as the source address in any packets received.

Next

pfSense is developed and maintained by Netgate. © ESF 2004 - 2024 View license.

Right Ctrl


ubuntu(Internal web server) (Linked Base for ubuntu(Internal web server) and Company router(Ubuntu)) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities
Firefox Web Browser
Mac 17 10:33
pfSense.home.arpa - Ser
https://10.0.2.8/services_snmp.php
pfsense COMMUNITY EDITION
System Interfaces Firewall Services VPN Status Diagnostics Help
WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
Services / SNMP
SNMP Daemon
Enable
Enable the SNMP Daemon and its controls
SNMP Daemon Settings
Polling Port
161
Enter the port to accept polling events on (default 161).
System Location
System Contact
Read Community String
public
The community string is like a password, restricting access to querying SNMP to hosts knowing the community string. Use a strong value here to protect from unauthorized information disclosure.
SNMP Traps Enable
Enable
Enable the SNMP Trap and its controls
SNMP Modules
SNMP modules
MibII
Netgraph
PF
Host Resources
UCD
Right Ctrl

pfsense [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

```
 8) Shell


Enter an option:

FreeBSD/amd64 (pfSense.localdomain) (ttyv0)

*** Welcome to pfSense 2.3.4-RELEASE (amd64 full-install) on pfSense ***

 WAN (wan)       -> em0        -> v4: 172.16.1.1/24
 LAN (lan)       -> em1        -> v4: 172.16.2.1/24
 OPT1 (opt1)     -> em2        -> v4: 172.16.3.1/24

 0) Logout (SSH only)                  9) pfTop
 1) Assign Interfaces                 10) Filter Logs
 2) Set interface(s) IP address       11) Restart webConfigurator
 3) Reset webConfigurator password    12) PHP shell + pfSense tools
 4) Reset to factory defaults         13) Update from console
 5) Reboot system                     14) Enable Secure Shell (sshd)
 6) Halt system                       15) Restore recent configuration
 7) Ping host                         16) Restart PHP-FPM
 8) Shell


Enter an option:
```

Right Ctrl

Part 1

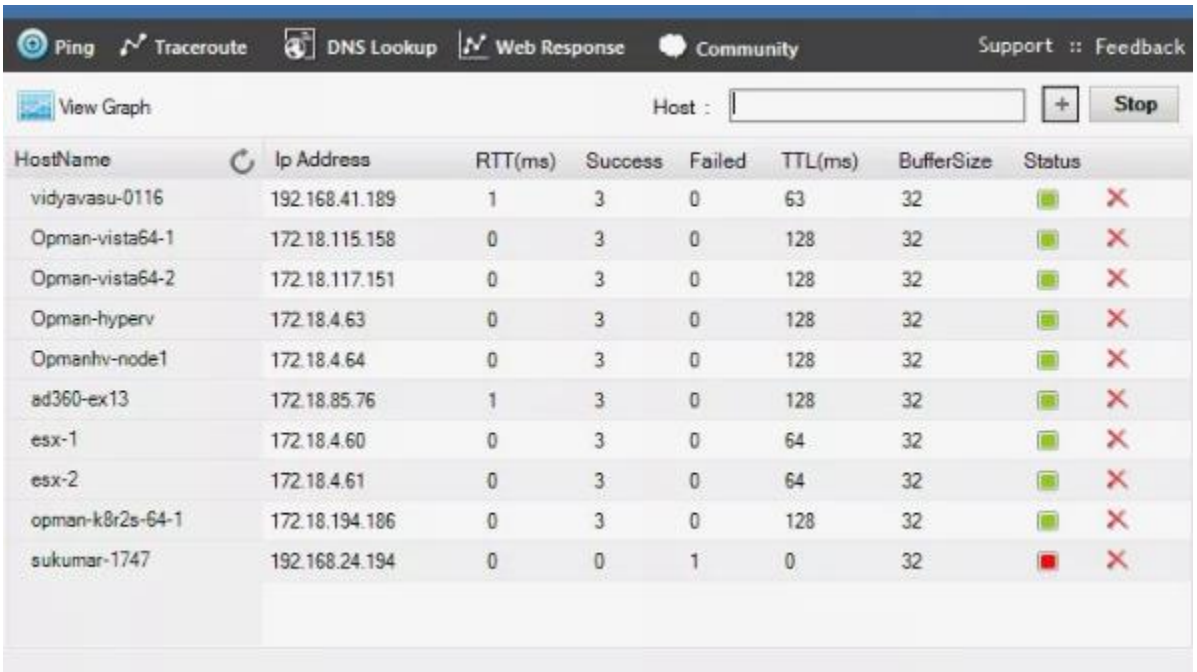
Network Monitoring

1. Commands

A number of commands are used to carry out network monitoring. The next section will offer comprehensive commands that are commonly used with screenshot on how they are implemented.

1.1. Ping

In general, if there is a problem with a particular network device (for instance, if a computer cannot connect to a server), Ping can be used to find where the problem may lie. If the server itself cannot be pinged, the issue is often a faulty network infrastructure, such as a hub, a switch, or a router (Elahi & Cushman, 2023). If the server can be pinged but the client machine cannot, this suggests that the client has some connectivity problems. For example, the command "ping -n 3" followed by a destination IP address will send a sequence of three ICMP Echo requests. By examining the responses that are returned, an administrator can get an indication of where the connectivity problems lie. The ping screenshot is shown below



HostName	Ip Address	RTT(ms)	Success	Failed	TTL(ms)	BufferSize	Status
vidyavasu-0116	192.168.41.189	1	3	0	63	32	
Opman-vista64-1	172.18.115.158	0	3	0	128	32	
Opman-vista64-2	172.18.117.151	0	3	0	128	32	
Opman-hyperv	172.18.4.63	0	3	0	128	32	
Opmanhv-node1	172.18.4.64	0	3	0	128	32	
ad360-ex13	172.18.85.76	1	3	0	128	32	
esx-1	172.18.4.60	0	3	0	64	32	
esx-2	172.18.4.61	0	3	0	64	32	
opman-k8r2s-64-1	172.18.194.186	0	3	0	128	32	
sukumar-1747	192.168.24.194	0	0	1	0	32	

1.2. Traceroute

The output of Traceroute shows a list of routers in the path from our broadband provider to the target IP of 8.8.8.8, along with the time taken for the signal to travel, measured in milliseconds.

It's common to see three entries for each router in the format (value1 value2 value3). These are the response times for three consecutive echo request packets sent to the router (Lee et al., 2014). The value given is the round-trip time for each packet in the order they are sent. One use for Traceroute is to identify where a connection to a service may be slowing down or failing.

traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets 1 192.168.1.254 (192.168.1.254) 4.126 ms 3.445 ms 6.647 ms 2 lo0-central10.pcl-ag04.plus.net (195.166.128.140) 20.308 ms 20.495 ms 20.327 ms 3 link-a-central10.pcl-gw02.plus.net (84.93.232.24) 21.398 ms 22.610 ms 22.773 ms 4 72.14.212.60 (72.14.212.60) 22.441 ms 23.479 ms 22.377 ms 5 66.249.94.234 (66.249.94.234) 24.497 ms 24.581 ms 26.858 ms 6 64.233.174.55 (64.233.174.55) 26.704 ms 64.233.174.169 (64.233.174.169) 26.886 ms 64.233.174.55 (64.233.174.55) 26.875 ms 7 108.170.248.241 (108.170.248.241) 27.797 ms 108.170.248.95 (108.170.248.95) 29.983 ms 108.170.248.241 (108.170.248.241) 28.471 ms 8 * * * 9 google-public-dns-a.google.com (8.8.8.8) 33.086 ms 29.329 ms 30.918 ms

1.3. Netstat

Netstat provides information and statistics about protocols in use and current TCP/IP network connections. It is a common first step in diagnosing network issues; by displaying all current TCP/IP network connections, Netstat can provide an initial overview of possible problems. With no additional parameters, current TCP/IP network connections are displayed. In addition, Netstat can be used with a variety of parameters to customize its output. The list of supported parameters is as follows: - e: display Ethernet statistics - s: display per protocol statistics. By default, statistics are shown for IP, IPv6, ICMP, ICMPv6, TCP, TCPv6, UDP, and UDPv6 - P: display the process ID and the name of the process to which each socket belongs - r: display the routing table - an: shows all connections and listening ports - o: displays active TCP connections and includes the process ID for each connection. This can be helpful in identifying the origin of a network service that may be the subject of a cybersecurity attack.

1.4. Nmap

Nmap is a highly effective network scanner which allows the determination of open ports and services on a computer. It can be useful for determining the ports that are open by checking the status of the firewall. Hence, its use is common. The Nmap command lines are as simple as nmap -san -vv target, where -sA is a scanning option and -vv is for additional verbosity. The most common scanning flag '-s' is given it helps user to describe kind of scan. For example: "-sT" describes a TCP connect scan that checks if a given TCP port is up and whether connections are accepted. Additionally to the "-sS" flag that indicates a TCP one, "-sU" tells that the user runs a UDP probe. With "-F" you can perform fast scan which include all the files in the target folder. This is to fmap which scan just only 100 most common ports.

2. Configuration Files

The main configuration of syslog is /etc/syslog.conf. When a system message is created, then this rule is present in the file and process the message. Each rule defines the facility, the severity level and the action to be taken by the designated personnel, in case a message matches it. So,

you will find another global log file with the name /var/log/syslog as well, which is a host for messages that match the conditions in syslog.conf. Every line within a respective config file is represented as a single rule. Lines of empty space and those starting with a hash symbol are just ignored, which means it's possible to split up the configuration as well as add comments for them by these signs. The syslog file has three basic keywords: syslog.conf file config suggests ["log", "equal", "action"] lets in "log" entry, then invokes ["log", "equal", "action"] to take a log level. The log level specifies what kinds of messages can be entered into log file. The available log levels from highest to lowest priority are: Severity: (0) "emerg", (1) "alert", (2) "crit", (3) "err", (4) "warning (5) "notice", (6) "info", and (7) "debug". The threshold of the notification level will typically be the "notice," in which cases, logs will be recorded only for message types of "notice" and above.

3. Monitoring Tools

WireShark is an open source, free, and widely used packet sniffing tool. Such program is used for collection, viewing and analyzing the data frames and the traffic, as they travel over the network. Capture and analysis of data traffic might have a rightful place within the scope of network monitoring. However, it can also be used by the hackers to observe or to eavesdrop on passages on the network.

The main menu of Wireshark also has a line named "Capture". Decide on interface from the choices on drop-down menu generated by this option. Opening a new window in Wireshark exposes the list of network interfaces with information such as description, type etc, An interface card for instance, say an Ethernet card or a network card, might be recognized as "Wireless Network Connection" and include some details.

4. Firewalls

Firewalls are software programs or hardware devices that filter and control the flow of data that is passed between a network and the outside world, as well as data that is passed between different areas within a network. Networks with active internet connections are probed at least once a day, and websites are probed every three minutes. Firewalls provide a first line of defense against attacks from outside the network. All traffic from the outside and from the internet has to pass through this powerful "barrier" before it can reach the internal computer systems. Hardware firewalls can be purchased as a stand-alone product, but many routers now come with a firewall built into them, which provides a very cost-effective level of protection. Hardware firewalls can be effective with little or no configuration, and they can protect every machine on a local network. Most of the home network routers produced on the market today include the type of firewall that helps secure the network by allowing configuration of the router to pass the suitable traffic in and out of the network. With a growing number of internet-enabled devices continuing to grow, having an effective firewall in place is key to providing adequate security in both personal and professional settings. The two commonly used firewalls include Cisco ASA and Fortinet Fortigate

Cisco ASA (Adaptive Security Appliance) is a security device that combines firewall, intrusion prevention, and Virtual Private Network (VPN) capabilities. From a network monitoring

perspective, the three most important configuration steps are to configure logging on the device, to ensure that it is transmitting log data to the Syslog server, and to configure the monitoring tool to receive syslog messages from the device (Bhuse & James, 2020).

5. Debugging

Debugging is the routine process of locating and removing computer program bugs, errors, or abnormalities (Bhuse & James, 2020). The method of debugging varies greatly depending on the operating systems and programming languages involved. First, one should have a clear understanding of the process and system used.

5.1. Packet Sniffing

Packet sniffing through the use of network monitoring equipment and software is a very important process that is used to analyze the flow of data in a network. Although packet sniffing can be used for non-invasive purposes, such as analyzing the status of a network's security, the reality is that it can also be used to analyze the data that is being transmitted across the network itself.

5.2. Log Analysis

Log files provide an important source of information in network monitoring. Logs record valuable information such as what the system's doing and who's been trying to access it. Log analysis is the process of examining these logs to identify possible issues and maintain security. Logs are generated by various network devices, including servers, routers, switches, and firewalls (Elezi & Karras, 2022).

5.3. Flow Analysis

For instance, continuous monitoring and analysis of network flow may reveal a sudden surge in traffic data between certain devices at a particular time of the day, which is not expected based on the normal operation of the organization. Such patterns could be an indication of security breaches or potential malfunctions of network devices. Therefore, flow analysis plays a crucial role in the cybersecurity defense line as well, not just limited to network management and performance optimization. In fact, flow analysis can be utilized as an effective method to detect anomalous activities and threats in a network. Modern flow analysis tools often integrate functions that cater for cybersecurity professionals.

6. IDS (Intrusion Detection Systems)

Intrusion Detection System (IDS) is a technology used to automate the process of monitoring events occurring in a computer system or network, analyzing them for signs of security problems. They come in two major categories. The first one is the Network-based Intrusion Detection System (NIDS). Snort is an open-source NIDS that is widely used.

Snort works by capturing packets, parsing their headers and passing them through its detection engine (Tian & Gao, 2023). The detection engine performs protocol identification, IP defragmentation and, for stateful protocols, maintains per-stream state.

Cisco Firepower is an intrusion detection and prevention system that can be used to actively protect a network from unauthorized access. Firepower uses Snort as the intrusion detection engine. Unlike with Snort, where we monitor traffic, in the case of Firepower monitor access to the network, hence the term 'intrusion prevention'.

7. Port Mirroring

Port mirroring allows the administrator to mirror the packets entering or exiting several switch ports to a specific switch port (Bhuse & James, 2020). As a result, an intrusion detection system sensor connected to the mirrored port can see all the packets on the network. With port mirroring, the administrator sees an intrusion in real time, typically as an alarm in an intrusion detection system sensor application.

Network Management

1. Quality of Service (QoS)

Quality of Service (QoS) is a set of techniques to manage network resources. In the context of QoS, quality refers to a measure of the network's performance as perceived by the users, and service refers to the manner in which the network provides resources to accomplish a task (Aboubakar et al., 2022). Basically, QoS is to provide an acceptable quality to different users and different application programs, which mean different users and different applications could have different treatments from the network, and the network resources are allocated and shared based on different QoS requirement of different users and different applications.

1.1. Traffic prioritization

The goal of traffic prioritization is to enhance the network performance by giving priority to certain types of traffic that are more important, in terms of latency or bandwidth, over the other types of traffic. Normally, we should prioritize the following types of traffic, from the highest to the lowest: Voice - Video - Critical Data - Regular Data - Bulk Traffic (like large file transfer). By default, the routers and the switches treat all the traffic equally.

1.2. Bandwidth allocation

Bandwidth allocation is the most important and commonly used feature of QoS. It is the technique of giving the best priority of the bandwidth to specific types of traffic. Bandwidth allocation allows a network administrator to reserve bandwidth for critical applications and to specify the minimum acceptable performance level for selected traffic (Keshari et al., 2021). Without using bandwidth allocation, any device on a network can send any level of traffic at any time, causing network congestion and increasing latency.

1.3. Packet scheduling

Packet scheduling can prevent network traffic jams and make sure that all the data can be transmitted more efficiently. Packet scheduling works at the data link layer of the OSI model, which controls how the packets are forwarded between the network. When a packet is received, it is stored in a buffer, and it will be transmitted to the next network device based on the output link scheduling algorithm. Nowadays, most network devices are using a more advanced packet scheduling algorithm called WFQ.

2. Load Balancing

Network traffic does not always flow in consistent patterns, and at times a server may experience overloading while another server remains underutilized. The objective of load balancers is to provide a clever way of sharing the workload across multiple servers. By default, when a client wants to access a certain service, it will send a request to the server configured as the default server.

3. Redundancy Protocols

In today's enterprise network, both STP and VRRP play a critical role in providing highly available networking environment. These protocols are used extensively by network engineers and system administrators to ensure network up-time and to provide seamless failover capabilities (Easttom II, 2023). By examining the features of STP and VRRP, we can see that they are designed to prevent network meltdown and to reconfigure the network quickly when there's a network outage.

4. Granular Management

Granular management is achieved through the use of Simple Network Management Protocol (SNMP) and Netflow. SNMP is a widely used protocol for this purpose in Granular Management. It supports variables in different data types, can manage and monitor all standards-compliant network devices and is particularly useful for a network with a lot of similar devices. Most importantly, it allows to set a remote configuration and get the configuration settings of the remote device and also to receive a trap message to a certain event (Kirkbride & Kirkbride, 2020). On the other hand, a concept of NetFlow has been introduced which gives the ability to collect records about all the data flows and to report on some specific fields - like number of bytes, packets, source and destination IP addresses etc. NetFlow data can be used by both IT professionals and business. For example, IT professionals can use this data to identify the cause of congestion in network whereas the Marketing department may use this kind of data to find out the popular sections of the company portal.

5. Layer 2 Techniques

Mostly deployed in LAN environments, VLAN separates the physical network into several different logical networks. Different departments, different project teams or even different virtual machines can be separated from each other by VLAN, regardless of their physical locations

(Svoboda et al., 2024). VLAN membership can be configured through software instead of physically relocating devices or connections. It can also reduce the workload on network administrators. We need to configure VLAN on the switch port and then connect the devices to the port. You should be able to install a VLAN capable network interface card to servers, workstations, router, etc. and configure the VLAN in the operating system..

5.1. VLAN Configuration

VLANs, or virtual local area networks, are a capability in the 802.1Q standard which allow a network to be set up so as to virtually create multiple, independent networks, all sharing the same network hardware. This is particularly useful if you have a larger network which needs to be 'segmented' - for example, in a school a network could be split into different VLANs for administration, for English, for Science and so forth. Each VLAN functions as a separate, independent network, even though in reality the devices in that VLAN can be spread over a number of different switches and network cables.

5.2. Port Mirroring

Port mirroring is a common feature and is available on most of the enterprise class and prosumer switches. The basic steps to set up port mirroring are as follows. First, the chosen port (which receives the mirrored packets) must be enabled for port mirroring. Then, the port(s) whose traffic will be monitored need to be selected (Pluskal et al., 2020). This is usually a port connected to a particular computer or networked device. The final step is to specify which port will receive the mirrored packets. It is usually the port that you’re monitoring or sniffing software is using.

6. Layer 3 Techniques

Layer 3 techniques involve routing and managing network traffic at the network layer. The section includes protocols like OSPF Routing Protocol and BGP Routing Protocol, which help in efficient routing and addressing. Open Shortest Path First (OSPF) is a link-state routing protocol that is widely used. It calculates the shortest path to establish the routing table.

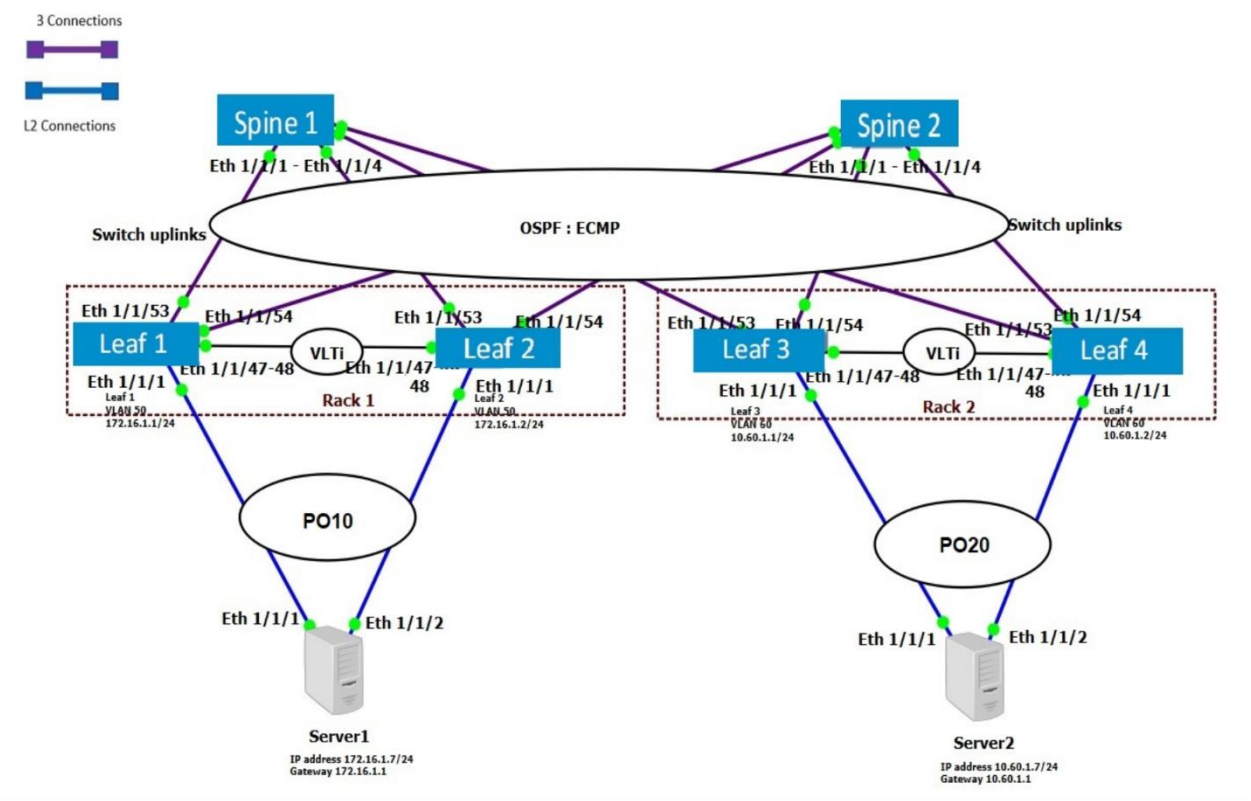


Figure 1: OSFP

When a data packet arrives at a router, the router must decide where to send that packet next. This decision is based on the information in the router's routing table. In a simple router, all the routing information is put into the routing table by network managers, but in a dynamic routing environment, such as in large networks or the internet, routing tables are built by the router itself using a routing protocol.

6.2. BGP Routing Protocol

The BGP (Border Gateway Protocol) is a routing protocol used to route traffic between different networks on the internet. Unlike internal gateway protocols such as OSPF, which are used to route traffic within a single autonomous system, BGP is designed to exchange routing information for the internet. One of the key features of BGP is its ability to consider many different paths for routing and to choose the best path based on a number of criteria. This is in contrast to other common gateway protocols such as the Open Shortest Path First (OSPF) protocol, which is based on the shortest path algorithm and therefore will only ever find the route from a source to a destination that has the least number of 'hops', or routers along the way.

Network Automation

1. Purpose and Application of Network Automation

Automation is crucial for two main reasons. First, it helps in identifying devices in the network and collect their properties for further usage. Network diagram is a visual illustration of how different devices are connected. When discovery is finished, we will get visual diagrams to demonstrate the network. By using network discovery and diagram software, we are able to fully automate infrastructure discovery and mapping. In addition, real-time information could be collected and presented on the diagrams. This visualized information could give network administrators a quick and direct view of how the network is structured and how different components are connected.

Second, it fosters configuration management. Configuration of devices could be standardized and harmonized. Consistency and conformity can be achieved so that misconfiguration and human mistakes can be minimized. For instance, a change carried out by a technician affects a device, this change will be recorded and configuration files on this device can be compared with those before the change.

2. Samples of Commands, Configuration Files, and Deployment Tools

In the field of network automation, there are many different tools and technologies available that can be used to automate the process of deploying a network configuration or a change across a large number of networking devices. One of the most commonly used tools is "Automation with Ansible". Ansible is an open-source software platform for configuring and managing computers, for the purpose of providing some sort of software deployment, and in our case, network configuration with zero downtime.

Output of the sample Python code is shown below, which demonstrates the successful retrieval of the defined route from the device and its corresponding network towards the destination IP address 192.12.126. This is a very simple but practical example to show how Python can be used to automate network processes by sending and receiving terminal commands to the networking devices through SSH channel.

```
# Disconnect the SSH session
client.close()
```

```
# Use the Paramiko SSH client to execute SSH commands
connected_shell = client.invoke_shell()
connected_shell.send('enable\n')
time.sleep(1)
connected_shell.send('cisco\n')
time.sleep(1)
connected_shell.send('terminal length 0\n')
time.sleep(1)
connected_shell.send('show ip route\n')
time.sleep(2) # Wait for the device to send back the output
output_from_device = connected_shell.recv(65535)
print(output_from_device.decode('utf-8'))
```

```
# Set the parameters to connect to a networking device using SSH
client.set_missing_host_key_policy(paramiko.AutoAddPolicy())
client.connect('192.168.8.1', '22', 'admin', 'cisco', look_for_keys=False, allow_agent=False)

# Create a new SSH client object
client = paramiko.SSHClient()
```

```python
# import the required libraries
import paramiko
import time
```

3. Management, Data, and Control Planes in Network Automation

One of the key tasks performed in network automation is data preparation. Data preparation can involve a number of responsibilities including data collection, data entry, and data cleaning. In the context of network automation, data preparation involves extracting information about the network's current state from the network via APIs or command line queries and storing this information ready to be arbitrated upon. Arbitration is the process of selecting a course of action based on given information. For automation to be effective, modern networking adopts element/system-centric views to the network rather than traditional views of a network. In essence, for automation to be effective, modern networks see the network as a series of interconnected systems which are to be managed as part of data center automation or what is more commonly referred to as DevOps. Traditional networks are managed and monitored in isolation on a platform-centric basis using a management platform or element manager which can be categorized as either command line editing or GUI-based systems.

4. Considerations for Automating OS Deployments across a Network

IT departments in medium to large organizations are faced with a dilemma - massive repetitive workloads that are time-consuming and error-prone. Each time a new server is deployed, the OS installation needs to be performed, and the server needs to be configured according to the function it will perform. Even more testing and troubleshooting is required if we add that server into a big network environment. That is where OS deployment automation comes in. Different from automating OS installation for a standalone server, one must also consider automating post-installation activities and network-related configurations. In this case, a good OS deployment tool can help. Therefore, on top of being able to perform automation on OS installations for multiple servers, the chosen tool must also support scripting to automate post-installation activities, tendency to perform unattended installation without user's intervention, and capability to customize installation and partition settings. In addition, as the network grows, managing these IPs to avoid conflict will become a burden. Hence, the automation solution must also consider how NIC settings and network domain to be configured automatically and consistently. Last but not the least, the chosen tool should be able to handle different types of servers, including both physical and virtual machines. Nowadays most of the servers have dual power supplies, especially for those critical servers that require zero downtime. When one of the power

supplies failed, the server will not be shut down and continue running with the other power supply.

Part 2

Setting up the Company Router

To set up the company router, go to an area where you can connect to the internet. Take one end of the yellow Ethernet cable and plug it into the 'Ethernet' port on the router. This is the port closest to the power socket. Plug the other end of the yellow Ethernet cable into the 'Ethernet' port on the intake socket. This is in the wall and is where the broadband cables from the hub join up. You can now check that you can access the internet on your computer. Connect the router to a power socket using the supplied power lead and adapter. The router will take a few minutes to power up. When it is plugged in and ready to go, the power light should be green.

Configuring the External Web Server

First, we need to log into the account provided by the domain name registrar, and then it is a simple matter of configuring the DNS settings and entering the IP address of the DNS server. With the changes saved, the domain name will now be connected to the configured DNS server. This change may take a couple of hours to fully process. This period is known as DNS propagation. Propagation is the time it takes for changes made to the domain name to be updated across the internet. Subsequently, when the domain name is typed into a web browser, the specified web page hosted by the external web server will appear. After the domain name and the DNS server are successfully configured, an IP address and a subnet mask need to be assigned to the external network adapter of the server. The IP address is used to identify the server among the hosts in the network. It is unique, meaning that no two hosts have the same IP address in the same subnet. The subnet mask helps to determine the relationship between the IP address of the computer and the hosts in the local network. To assign the IP address, open the Network and Sharing Center from the Control Panel and click on Change adapter settings. Then, right click on the Local Area Connection adapter and click on Properties. In the networking tab, double click on Internet Protocol Version 4 (TCP/IPv4). Choose the option "Obtain the following IP address" and enter the IP address and the subnet mask. Click OK to apply the settings. By following the above steps, the external web server should be up and running, accessible through the configured domain name.

Establishing Communication with the External Client PC

After setting up the external web server, it is important to establish communication between the network and the server to facilitate data transfer and remote accessibility. Firstly, the internet port of the physical router was connected to the first port of the wireless router with an Ethernet cable. The internet port is where the Wide Area Network or WAN cable should be connected. The configuration page of the wireless router can be accessed by typing the default gateway IP address on the address bar of a web browser. The default IP address and the login credentials are usually printed at the back of the wireless router. Once managed to get access to the

configuration page, a static IP address of the first port of the wireless router should be set in a way that is in the same subnet as the internet port IP address of the company router.

Implementing Kiwi Syslog for Network Monitoring

To provide network monitoring service, a network management system takes the help of Simple Network Management Protocol (SNMP). In this demonstration, the Kiwi Syslog Server is used, which provides a 5-device free version and is able to receive syslog messages sent by the router. First, in the router web configuration page, at "logging" -> "logging host", need to input the Kiwi Syslog Server's IP address and it will start to send syslog messages from the router to the server. Then in the Kiwi Syslog Server window, click "File" -> "Setup" and at "Inputs" tab, click "Add new" to add a new device. The device type should be "syslog" and at "IP address", input the router's IP address. The next step is to add a log file to store the syslog messages. Go to "File" -> "Setup" -> "Kiwi Syslog Daemon" tab and at "LogFile" option, click on the empty small box and select a location to save the log file. In Kiwi Syslog Server, it is crucial to customize the number of the log files and the maximum size of each log file, which is 20. So, it will allow storing up to the latest 20 archives of the log file. At the "Device" tab, one can change the display name of the router and choose to enable or disable the storing of the syslog messages to a log file. When a network uses Kiwi Syslog Server for network log management and analysis, the server can monitor IP traffic generated by syslog messages of an IP device. Each log message contains an indication of the type of message and the contents of the message.

Enabling Communication between the Internal Web Server, Staff PC 1, and Observium

After ensuring the correct configuration of the web server, the next step is to install Observium on the Ubuntu virtual machine to see the monitoring of the network devices. First of all, we are updating the system by using the Update command. This will retrieve the newest version of the packages. To do this, the terminal is opened in the Ubuntu virtual machine and we write a command "sudo su" to log in as a root user. Then we write "apt-get update" to update the system. After the completion of updating we have to install the "SNMP" and "add SNMP mibs". SNMP is required for Observium. Then we need to write the command to install SNMP and press "Y" and then hit enter to continue with the installation. Subsequently, enter the command to add the SNMP mibs. After completing the required steps, we need to populate the Observium database. But before populating the database, we have to install the "composer" and "required PHP modules". "Composer" is required for the series of package installations. Initially issue the command to install the "required packages" and then press "Y" and then hit enter to continue with the installation. Subsequently write the command to install the "composer" and then press "Y" and then enter to continue with the installation. Then we have to move to the "/opt" directory and install the "composer" by using the commands in the terminal. First of all write "php -r ", then "copy('https://getcomposer.org/installer', 'composer-setup.php');" and "php -r ". Enabling the communication of the internal web server, staff PC 1 and Observium can be categorized into the following 4 important steps where multiple tasks will be carried out.

Configuring PfSense to Block External Traffic

Step 1: Create an alias containing the IP addresses of the internal DNS servers. On the PfSense web interface, go to "Firewall" and then "Aliases". Click on "Add" to add a new alias. Give the alias a name, for example "Internal DNS IPs" and then add the IP addresses of the internal DNS servers.

Step 2: Create a rule to block traffic from the internal network to the outside. Still, on the PfSense web interface, go to "Firewall" and then "Rules" and then "LAN". Click on "Add" to add a new rule. Make sure the following are selected:

- Action: Block

- Interface: LAN

- Address Family: IPv4

- Protocol: any

- Source: any

- Destination: not: Internal DNS IPs

- Log: ticked

- Description: any

Click on "Save" to save the rule. Now we should see the rule in the rule list. When the rule is applied, the internal network will only be able to access the web but nothing else. If set, the rule will log every packet it denies.

Step 3: Create an alias containing the external IP addresses that we do not want to have any direct access to the internal network. On the PfSense web interface, go to "Firewall" and then "Aliases". Click on "Add" to add a new alias. Give the alias a name, for example, "External Not Allowed". Next, add the IP address of the external network.

Step 4: Create a rule to block traffic from the external network to the internal network. The only difference between this step and step 2 is in the destination. When applying the rule, make sure to change the destination to the "External Not Allowed" alias. This will prevent any traffic from the external network, except for the responses to the requests initiated by the internal network, to go through to the internal network.

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